RAGX System Architecture & How It Works

A Teammate's Guide to Data Quality, RAG Ingestion, and Hallucination Detection

1. What is RAGX and Why Do We Need It?

Modern Artificial Intelligence (AI) uses Large Language Models (LLMs) to answer user questions. However, raw LLMs often suffer from a critical flaw called **Hallucination**—they generate incorrect facts with absolute confidence.

Retrieval-Augmented Generation (RAG) solves part of this problem by giving the LLM internal company documents (PDFs) to read before answering. However, standard RAG systems still fail when documents contain errors or when the AI mixes up facts.

RAGX is our end-to-end platform that audits document quality and continuously evaluates answer reliability before answers reach the user.

> ***Key Takeaway for Team:*** *RAGX ensures that every AI answer is 100% traceable back to specific lines and pages in our uploaded PDFs. If an answer cannot be proven by our documents, RAGX catches and flags it immediately.*

2. How RAGX Works: The 3 Core Phases

RAGX operates in three distinct, automated phases to transform raw PDFs into verified, hallucination-free answers:

Phase	Name & Purpose	Key Output
Phase 1	**RAG Data Ingestion & Retrieval** Parses uploaded PDFs, converts text into mathematical vector embeddings, and stores them in ChromaDB.	Indexed Vector Chunks & Candidate Contexts
Phase 2	**Data Quality Audit Engine** Audits uploaded documents before querying. Checks text completeness, chunk diversity, and knowledge contradictions.	Health Score & Contradiction Matrix
Phase 3	**Answer Reliability & Hallucination Evaluator** Decomposes LLM answers into atomic claims, matches claims to evidence, and computes composite reliability score (S_Ans).	5-Tuple Traceability & Reliability Badge

3. Step-by-Step Data Journey (From User Query to Verified Answer)

- **Step 1: Document Upload & Chunking**: When a PDF (e.g. *2ND SEM RESULT.pdf*) is uploaded via the **Documents Tab**, PyMuPDF extracts text and PyTorch splits text into 500-character vector chunks.

- **Step 2: Vector Store Indexing**: Each chunk is embedded using the all-MiniLM-L6-v2 neural model and indexed in ChromaDB vector database.
- **Step 3: User Query Submission**: When a user asks a question (e.g. *'What are the marks in Mathematics-II?'*), ChromaDB searches for top-3 semantically relevant chunks.
- **Step 4: Answer Synthesis**: The RAG engine builds a context prompt and synthesizes a direct answer derived strictly from the retrieved chunks.
- **Step 5: Claim Extraction & Matching**: Phase 3 ClaimExtractor breaks the answer into atomic factual statements (e.g., *'BMATS201 MATHEMATICS-II 35 11 46 F'*) and verifies each against source evidence.
- **Step 6: Reliability Scoring & Badge**: AnswerEvaluator calculates composite reliability score (S_Ans) and displays a green **HIGHLY_RELIABLE** or red **UNRELIABLE** badge.

4. How RAGX Scores Answer Reliability (S_Ans)

RAGX does not guess reliability—it uses a mathematically deterministic formula bounded between 0% and 100%:

S_Ans = (0.60 × Claim Support) + (0.20 × Citation Coverage) + (0.20 × Retrieval Similarity)

Understanding 5-Tuple Traceability:

Every claim generated by RAGX is linked to a strict 5-tuple proof chain:

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Claim Text ■■> Evidence Text ■■> Chunk ID ■■> Document Name ■■> Page Number
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5. Teammate Guide: Using the RAGX Web App

Tab Name	What Teammates Can Do Here
■ Documents	Upload new PDFs, view indexed document status, preview PDFs directly in-web, and soft-delete/restore documents from trash.
■■ Data Quality	Audit uploaded PDFs for text extraction completeness, chunk diversity, and knowledge contradictions before running queries.
■ RAG Chat	Ask questions about company documents and receive instant answers backed by document chunk references.
■■ RAGX Evaluator	Run in-depth reliability reports on any query. View claim-by-claim breakdown, support status, and 5-tuple citation proofs.
■ Analytics	View system-wide evaluation metrics, average reliability scores over time, and risk distribution charts.

Summary: RAGX provides total confidence in AI answers by pairing dynamic vector retrieval with deterministic 5-tuple proof verification.